



Passionate and aspiring DevOps Engineer with a strong foundation in Python, cloud automation, and CI/CD pipelines. Skilled in Docker and Git, with hands-on experience in deploying scalable systems and optimizing workflows. Eager to apply my skills and customer-focused insights to drive innovative solutions in a DevOps role.

konnect.surabhi@gmail.com

Ritterstr. 16, 09111

<https://ssh-surabhi.netlify.app/>

Indian

Chemnitz, Sachsen

0176 83334794

Married

Social networks

@ssh-surabhi

@ssh-surabhi

@SaiSriHarsha

@live:surabhi.harsha

Languages

German
B2

English
C1

Telugu
Native [Mother Tongue]

Hindi
C1

Skills

AWS-DevOps | Python | Terraform | Kubernetes | Docker | Ansible | Jenkins | Gitops | Linux | BASH | PostgreSQL | MongoDB | Automation | Testing | Image Processing | PyQt | Machine Learning | Django | HTML | CSS | MATLAB | React | REST API | C++ |

Work experience

Development Engineer 06/2021 - 06/2024
Moewe Optical Solutions GmbH, Mittweida, SN, Germany

- **Algorithm Development:** Developed an error detection algorithm, using image processing methods for laser projection.
- **Desktop GUI:** Created a cross-functional desktop application using PyQt5 to support laser functionalities.
- **Automation:** Designed automated test scripts (pytest), including regression and unit tests, while working with I2C, UART, TCP/IP, UDP, and Ethernet protocols.
- **Feature Updation:** Updated the software with new functional test features following customer requirements.
- **Linux-Application Development:** Created Linux daemons within Petalinux environment to optimize system functionality.
- **Containerizing and Collaboration:** Containerized applications using Docker and utilized Git for code management.

* Python | PyQt | Linux | OpenCV | VS Code | TCP/IP | Ethernet | Pytest | Gitlab | Docker | MySQL

Student Research Assistant (HI-Wi) 04/2020 - 09/2020
Technische Universität Chemnitz, Chemnitz, SN, Germany

- Conducted comprehensive **testing and evaluation** of 3D hand pose estimation algorithms for accuracy and performance.
- **Tested the hand pose estimation** pipeline with diverse datasets, validating outputs against established ground truth values.
- **Evaluated** pixel (2D) and positional (3D) **accuracy** to ensure the reliability of hand pose estimation techniques.

* Python | Keras | OpenCV | VS Code | Ubuntu | Data Analysis | Visualization | VGG Annotator

Software Test Engineer 10/2012 - 09/2014
Caagnota Info Pvt. Ltd., Hyderabad, India

- **Test Plan:** Creating test plans based on the project requirements.
- **Test Case Development:** Designed test cases and executed Feature Testing, System Integration testing, and Regression Testing, to contribute overall software quality assurance.
- **Error Analysis:** Analyzed and documented error reports for failed test cases.
- **Team Management:** Coordinated team activities and managed project workflows.
- **Agile Workflow:** Worked within Agile testing methods.

SDLC | STLC | Manual Testing | Jira | Agile | SQL

Projects

Devops Project [Deploying Spring-PetClinic in AWS Cloud] 06/2024 - 10/2024
Datscientest.com, Paris, France

- Deployed and managed a scalable, high-availability infrastructure on **AWS**, leveraging services such as **EC2, S3, RDS, Route53, and Lambda**. [[Git Link](#)]
- Automated the provisioning of cloud resources using **Terraform** and AWS CloudFormation, significantly reduce deployment time, and streamlineing setup.
- Designed and implemented **CI/CD** pipelines using **Jenkins, Gitlab CI, and AgroCD** to automate the testing, building, and, deployment of applications in AWS.
- Managed containerized applications using **Docker**, and **Kubernetes** (EKS).
- Improved application security by implemented by **IAM** roles, configuring **security groups**, and utilizing AWS Systems manager for automated security patching.
- Developed and maintained monitoring and alerting solutions using AWS CloudWatch, **Prometheus**, and **Grafana** to ensure quick incident response.
- Implemented effective **disaster recovery** solutions across multiple AWS regions, significantly reducing downtime and minimizing risks of data loss.
- Built pipelines in **Jenkins**, incorporating **Unit testing** to ensure code reliability.

Skills

Network Protocols
TCP/IP, UDP, ETHERNET

Software Programming
Python, Matlab, HTML, CSS, C# (Basic), C/C++ (Basic)

Communication Protocols
CAN, LIN

Technical Tools

Cloud Tools
AWS Fargate, EKS, EC2, VPC, Lambda, ECR, Gitlab, Jenkins, Docker, Terraform, Ansible, Grafana, ALB, Prometheus, Datadog, Airflow, Github

Editors and Tools
VS Code, PyCharm, Jira, Visualstudio, Wireshark, Trello, MicrochipStudio, H-Term, Confluence, Arduino, Putty

Certifications

DevOps, Ansible, Terraform, Kubernetes, Jenkins, Docker, SQL, NoSQL, Gitlab, Linux-Administration, AWS Solution Architect [Udemy], Deutsch, ISTQB [Udemy] [\[Link for certifications\]](#)

Other Skills

Backend Programming

- Django [RestAPI]
- FastAPI
- Flask

Manual And Autoimation Testing with Python

- GUI Tesing
- API Testing
- Manual Testing
- Electronic Board level Testing

Operating Systems

Windows [10, 11]
Linux [Ubuntu, Fedora, Scientific Linux]

Interests

Hiking/Trekking, Cycling.
Learning New Technologies

Projects

Terraform [Deploying highly available application in AWS] 09/2024 - 10/2024
Datascientest.com, Paris, France

- Designed a highly available and scalable deüployment using Terraform, leveraging modular and reusable code for repeatability and efficiency. [\[Git Link\]](#)
- Configured a secure AWS environment with 1 **VPC**, multiple subnets (private and public), and NAT gateways to facilitate internet access for private resources.
- Implemented **auto-scaling groups** for EC2 instances hosting the application, to ensure performance and availability.
- Deployed **AWS RDS** across multiple availability zones to achieve database redundancy and reliability.
- Configured an Application Load Balancer (**ALB**) to evenly distribute traffic, ensure fault tolerance, and enable secure HTTPS (**TLS**) access for web servers.
- Deployed a **bastion host** in an auto-scaling group to provide a secure and scalable entry point for managing private resources.

Kubernetes Project 08/2024
Datascientest, Paris, France

- Designed and deployed Kubernetes-based microservices: FastAPI, PostgreSQL, and PGAdmin, with scalable and secure configurations. [\[Github Link\]](#)
- Deployments using YAML, **Helm** charts, and Kustomize, ensuring reusability.
- Configured HorizontalPodAutoscaler (**HPA**) to scale pods based on CPU utilization and provisioned a shared 10 Gi PersistentVolume for storage.
- Secured the application with **SSL certificates** and exposed it via **Ingress** controller.
- Prepared a **disaster recovery plan** and implemented ETCD database backups.

Instance Segmentation for Human Localization [Master Thesis] 04/2019 - 09/2019
Technische Universität Chemnitz, Chemnitz, SN, Germany

- Creating disjoint body annotations using coco-annotator for human body regions
- Detecting and localizing Humans as objects in images.
- Segmenting separate human body regions while occluding with other objects.
- Testing maximum possibility for the right regions of humans with original masks.

Education

Diploma in Devops Engineer 06/2024 - 10/2024
Datascientest [Sorbonne University], Paris, France
AWS, Terraform, Kubernetes, Docker, Jenkins, Gitops, Ansible, Linux-Admin, Grafana

Integrationskurs Deutsch B1 11/2020 - 04/2021
Initiativ Gruppe, München, BY, Germany

Masters in Information and Communication Systems 10/2014 - 09/2019
TU Chemnitz, Chemnitz, SN, Germany
Image Processing, Machine Learning, Deep Learning, Computer Vision, Digital Systems

Masters in Electronics 08/2010 - 06/2012
Osmania University, Hyderabad, TG, India